How the model is trained

DFU Thermography Screening Station (DFUSCAN-01) · eMathrix Technologies · October 28, 2026

This document explains how the station's image classifier is trained and put on the station, for the research team and anyone reviewing the method. Training happens on a separate x86 computer. The station only runs the finished model and never trains or compiles anything.

> **What the model does and does not do.** The classifier looks at the colour image of the soles and outputs a predicted class (`class_0` , `class_1` , …). It never assigns a Wagner grade: grades come only from a licensed clinician and are used as training labels. The left–right temperature asymmetry (the 2.2 °C flag) is not part of the model. It is plain arithmetic and is shown separately, side by side with the classifier output.

1. Overview

```
Station scans (colour_registered.png) + clinician labels
   │
   ▼  manifest.csv              one row per image: patient_id, image_path, label, source, capture_session
   ▼  data-quality report       counts per class and per patient, warnings; nothing dropped or rebalanced
   ▼  training.train            patient-grouped 5-fold cross-validation, then a final model on all data
   ▼  training.export           ONNX model + model_meta.json, checked against PyTorch
   ▼  training.hailo_compile    .hef for the Hailo-8L, quantised on station colour frames
   ▼  copy to the station       models/ folder, then the hardware test
```

2. The data

Where it comes from

- The research team collects scans on the station. Each scan folder has a registered colour image (`color_registered.png`), the same image the station's classifier sees.
- A licensed clinician assigns the label for each patient. The station never supplies labels.
- Patients are identified only by the anonymised reference code the research team supplies. No names or other identifiers are stored.

The manifest

Column	Meaning	Required
`patient_id`	Anonymised reference code, never a name	Yes
`image_path`	Path to the image, relative to the image folder	Yes
`label`	Label assigned by the clinician	Yes
`source`	Where the image came from, e.g. the station id	No
`capture_session`	Session the image belongs to	No

If any row is missing a required value, the whole manifest is rejected with the line numbers listed. Rows are never dropped silently.

Data-quality report

Before training, the script writes `data_quality.md`: images and patients per class, class imbalance, very small classes and missing metadata. **Nothing is dropped, imputed or rebalanced.** The number of patients is the real sample size. If a class has only a few patients, the report says so, and that limitation must be stated wherever results are reported.

3. Splitting: by patient, never by image

All images of one patient stay on the same side of every split. Every cross-validation fold is grouped by `patient_id`, and the code checks this both when the folds are built and again when each fold is used. Splitting images at random would let the model see the same feet in training and testing and would overstate its performance.

4. Training

Setting	Default	Notes
Network	MobileNetV2	Small enough for the Hailo-8L; starts from ImageNet weights
Layers trained	Final layer only	The pretrained feature layers are frozen; `--train-backbone` trains them too
Input	224 × 224 RGB	ImageNet normalisation, the same preprocessing code as the station
Cross-validation	5 folds, grouped by patient	`--folds`
Epochs / batch size	20 / 16	`--epochs`, `--batch-size`
Optimiser	Adam, learning rate 0.001	`--learning-rate`
Class imbalance	Inverse-frequency loss weights	Computed from the training side of each fold only. The data itself is not rebalanced.
Augmentation	Random horizontal flip, small brightness and contrast changes	Training side only; `--no-augment` turns it off
Random seed	2026	Runs are repeatable on the same machine and library versions

```
python -m training.train --manifest manifest.csv --image-root images/ --out runs/run1 --final
```

Outputs: `data_quality.md`; `cv_report.md` and `cv_report.json` (accuracy per fold, recall per class, confusion matrix and the number of patients in each fold); `final_model.pt`; and `label_map.json`, which maps `class_0`, `class_1`, … back to the clinician's labels. The label map stays with the research team and is not put on the station.

No accuracy target is built into the software. Accuracy belongs to the dataset and is reported as found.

5. Export and checks

```
python -m training.export --run runs/run1 --out models/
```

Writes `dfu_mobilenetv2.onnx` (fixed input 1 × 3 × 224 × 224) and `model_meta.json`. The export is kept only if ONNX Runtime and PyTorch agree within 0.0001 on test inputs. The station refuses any model whose class names read like a clinical grade.

6. Compiling for the Hailo-8L

```
python -m training.hailo_compile --onnx models/dfu_mobilenetv2.onnx \
    --metadata models/model_meta.json \
    --calibration-images path/to/station/captures --out models/dfu_mobilenetv2.hef
```

- Runs on x86-64 Linux with the Hailo Dataflow Compiler and 16 GB of RAM or more.
- Quantisation needs at least 64 colour frames **from this station**, not images from elsewhere.
- The normalisation is built into the `.hef`, so the station feeds it plain RGB.

7. Putting the model on the station

1. Copy `dfu_mobilenetv2.hef` and `model_meta.json` into `models/` on the station.
2. Confirm the accelerator: `hailortcli fw-control identify`.
3. Run the hardware test: `pytest -m hardware tests/test_inference.py`.

8. Interim model from public photographs

Until enough station scans exist, a first model (`public_v1`) can be trained on public colour photographs, only to show the classifier path working. It must be retrained on station scans.

- Labels are mapped with an exact table that the research team confirms. Any label not in the table is rejected and listed, never guessed.
- Byte-identical duplicates are removed and listed.
- Where a dataset documents no patient ids, near-duplicate and visually similar pictures are grouped together and treated as one patient.
- Whole groups (64 images or more) are set aside for Hailo quantisation.

> **Limitation, to be stated with any result from this model:** grouping public photographs is only a proxy for patient separation. Two different-looking photographs of one patient can land in different groups, so cross-validation figures on public data may be optimistic. Public photographs also differ from station scans in camera, lighting and view.

9. Privacy

Manifests, images and reports contain session data. They stay on the research team's computers, are never committed to git, and are never uploaded to any online service. The station itself runs offline.
